

PROJECT ASSIGNMENT 2

Group10-PA2 Use Case Document

Twelve detailed use cases with UML, edge cases, and end-to-end traceability

FIFA.com and Chess.com

Course	CSC13112 - UI/UX Design, FIT-HCMUS
Group	Group10
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1. Scope, actors, and notation

System	Primary actor	External systems	Use cases
FIFA Ticket Confidence Module	Ticket planner	FIFA tournament catalog; governed ticket-state feed; official ticketing partner; notification and identity services	F-UC01-F-UC06
Chess Guided Beginner Review Module	Beginner / returning learner	Game store; analysis service; explanation service; learning catalog	C-UC01-C-UC06

Notation: Solid lines are actor associations. Dashed arrows labeled <<include>> point to required reused behavior. Dashed arrows labeled <<extend>> point from optional/conditional behavior to its base use case. External systems remain outside the system boundary.

UC-UML-F FIFA Ticket Confidence Module

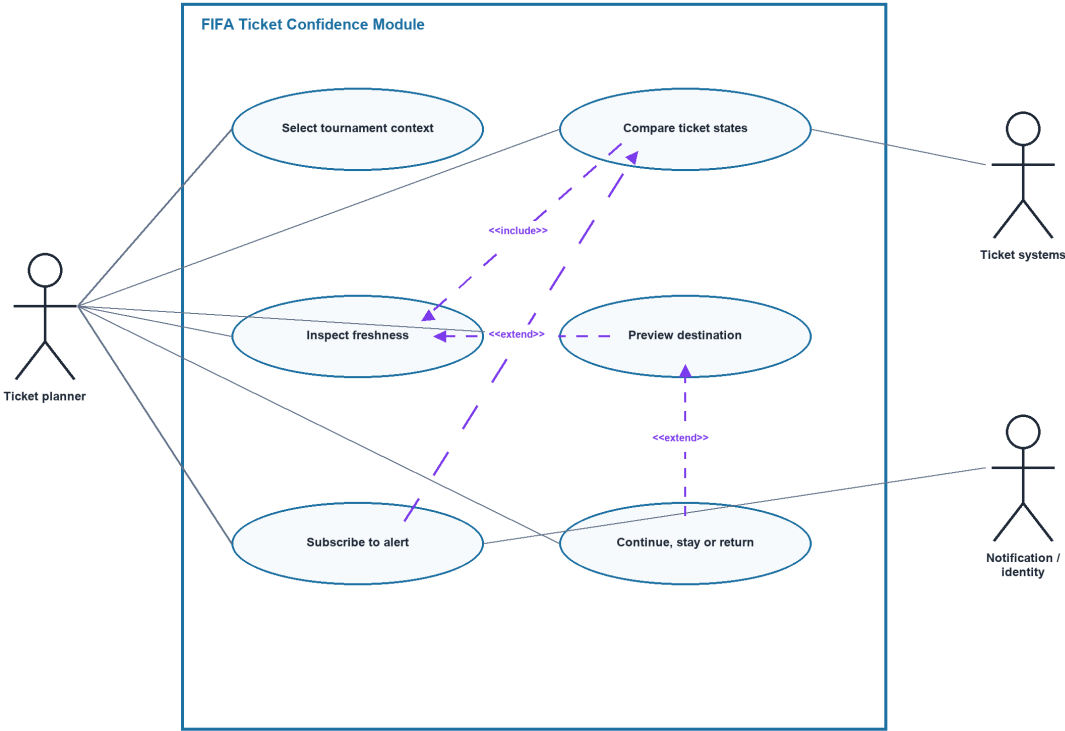


Figure UC-UML-01. FIFA UML use-case diagram
Source: Design inference; actors and systems derived from F-A1 and the traceability matrix
Related: F-UC01-F-UC06

UC-UML-C Chess Guided Beginner Review Module

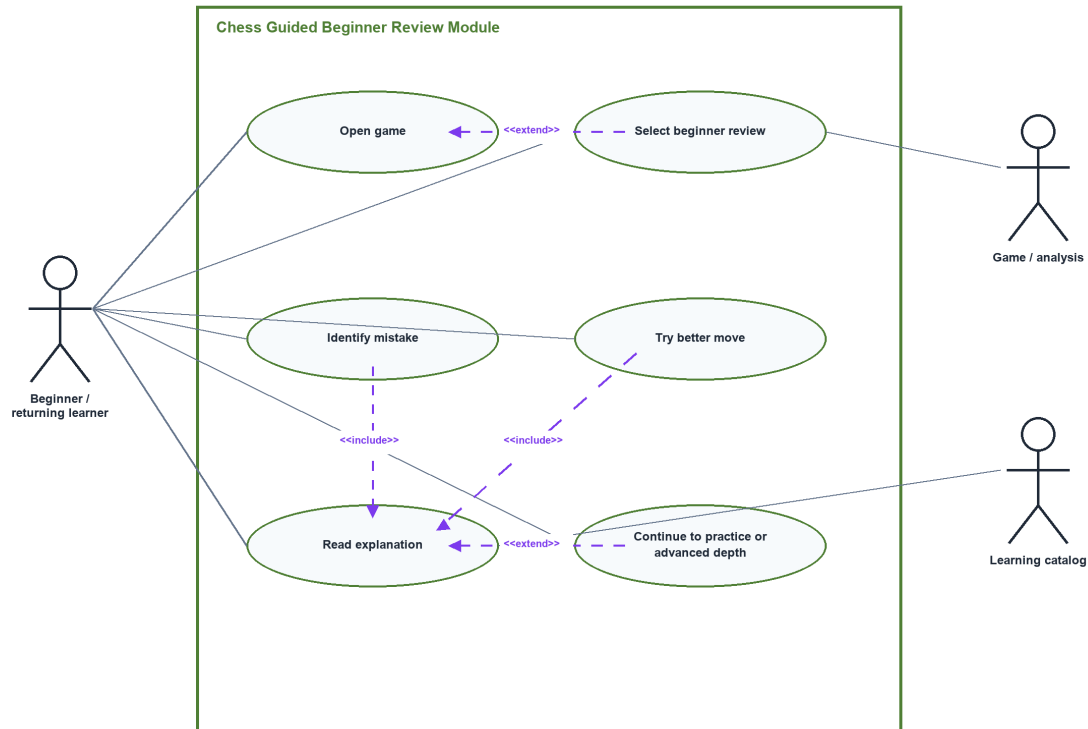


Figure UC-UML-02. Chess UML use-case diagram

Source: Design inference; actors and systems derived from C-A1 and the traceability matrix

Related: C-UC01-C-UC06

2. Detailed use cases

2.1 F-UC01 - Select tournament context

Field	Record
ID	F-UC01
Goal	Select tournament context
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	FIFA tournament catalog
Trigger	User opens the module from a tournament or ticket entry.
Preconditions	Module is available; a catalog response or safe cached list exists.
Success guarantee	A labeled tournament/team context is selected and preserved.
Priority	High
Frequency	Each new planning context

Main flow

1. Open the tournament selector.
2. Review searchable tournament and team labels.
3. Choose a tournament and optional team.
4. Confirm the selected context.
5. Load the ticket-state comparison for that context.

Alternate flows

- A1 at step 2: Catalog is unavailable; show the last timestamped list and a retry action.

- A2 at step 3: No context matches; clear filters and preserve the query.

Exception flows

- E1: Network loss retains the selection locally and labels the state offline.
- E2: Login mismatch offers account switch without exposing another account's plan.

Field	Record
Business rules	Tournament identity and freshness must be visible; no state is inferred from an absent catalog row.
Data	Tournament ID, team ID, locale, selected context, timestamp.
Usability	Recognition over recall; selected context stays visible.
Accessibility	Keyboard-operable search and selection; labels are announced.
Privacy	Selection may remain local until the user explicitly saves it.
Traceability	PA1 FIFA drawback -> F2-E03/E04/E09 -> F-SES-03/04 -> UR-P-F2 -> UR-F03 -> Mobile overview -> 1 vote -> TP-FIFA -> F-A1 -> PA3 context-resumption test.

2.2 F-UC02 - Compare ticket states

Field	Record
ID	F-UC02
Goal	Compare ticket states
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	Governed ticket-state feed
Trigger	A tournament context is selected.
Preconditions	Context is valid; state feed may return current, stale, missing, sold out, resale, waiting room, or hospitality states.
Success guarantee	Available states are compared without treating missing data as an outcome.
Priority	Critical
Frequency	Every planning visit

Main flow

1. Request states for the selected context.
2. Display each channel with a text state label.
3. Compare primary sale, resale, hospitality, wait, and interest options.
4. Explain unavailable and sold-out distinctions.
5. Choose a state to inspect or stay on the comparison.

Alternate flows

- A1 at step 2: State is stale; retain it with timestamp and warning.
- A2 at step 2: State is missing; show unknown and retry, not sold out.
- A3 at step 3: Waiting room is active; explain temporary queue behavior.

Exception flows

- E1: Feed conflict shows both governed sources and blocks a confident action.
- E2: Network loss retains the last labeled snapshot.

Field	Record
Business rules	Sold out, resale, waiting room, missing, and stale are mutually explicit labels.
Data	Context, channel, state, source, timestamp, eligibility note.
Usability	Visibility of system status and error prevention.
Accessibility	State uses text and icon, never color alone; table headers are semantic.
Privacy	No personal data required for comparison.
Traceability	PA1 FIFA ticket drawback -> F2-E09 -> F-SES-04/05/06 -> UR-P-F1 -> UR-F01 -> Ticket status clarity -> 10 votes -> TP-FIFA -> F-A1 -> recommendation -> PA3 state-accuracy test.

2.3 F-UC03 - Inspect freshness

Field	Record
ID	F-UC03
Goal	Inspect freshness
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	Governed ticket-state feed
Trigger	User opens freshness details for a ticket state.
Preconditions	A state record exists, even if stale or missing.
Success guarantee	Timestamp, source, and confidence boundary are visible.
Priority	Critical
Frequency	At important sale decisions

Main flow

1. Open freshness details.
2. Read last-updated time and source.
3. Compare current time with the freshness policy.
4. Review stale or missing explanation.
5. Refresh or return to comparison.

Alternate flows

- A1 at step 3: Record is stale; block unqualified action and offer refresh.
- A2 at step 4: No update exists; show unknown and escalation guidance.

Exception flows

- E1: Refresh fails; keep prior timestamp and recovery action.

Field	Record
Business rules	A refresh attempt does not erase a valid prior timestamp.
Data	State ID, observed time, source ID, freshness threshold.
Usability	Clear feedback and calibrated confidence.
Accessibility	Time is readable in local format and available to assistive technology.
Privacy	No additional personal data.
Traceability	PA1 evidence rules -> F2-E09 -> F-SES-04/06 -> UR-P-F1 -> UR-F01 -> Freshness and alerts -> 3 votes -> TP-FIFA -> F-A1 -> PA3 freshness test.

2.4 F-UC04 - Preview destination

Field	Record
ID	F-UC04
Goal	Preview destination
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	Official ticketing partner
Trigger	User selects an outbound ticket action.
Preconditions	Destination metadata exists or the action remains blocked safely.
Success guarantee	Domain, purpose, context transfer, and return behavior are previewed before departure.
Priority	Critical
Frequency	Before each outbound action

Main flow

1. Open destination preview.
2. Read official partner name, domain, and purpose.
3. Review what context will transfer.
4. Choose Continue, Stay, or Return.
5. If Continue is chosen, open the destination with a return route.

Alternate flows

- A1 at step 2: Metadata is partial; offer Stay and report the limitation.
- A2 at step 4: User stays; preserve comparison context.

Exception flows

- E1: Partner failure returns the user to the preserved state with an explanation.

Field	Record
Business rules	The module never claims a partner transaction is complete.
Data	Partner ID, domain, purpose, context token, return URI.
Usability	User control, trust calibration, and error prevention.
Accessibility	Destination and choices have explicit labels and logical focus order.
Privacy	Transfer only required context after consent; do not expose plan data in a URL.
Traceability	PA1 handoff drawback -> F2-E09 -> F-SES-02/05 -> UR-P-F1 -> UR-F02 -> Handoff trust -> 8 votes -> TP-FIFA -> F-A1 -> recommendation -> PA3 destination-recognition test.

2.5 F-UC05 - Subscribe to alert

Field	Record
ID	F-UC05
Goal	Subscribe to alert
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	Notification service
Trigger	User chooses to follow a state change.
Preconditions	A context is selected; permission is not assumed.
Success guarantee	Alert preference is saved or a clear non-notification alternative is provided.
Priority	High
Frequency	During wait/interest states

Main flow

1. Choose Subscribe to alert.
2. Select relevant state changes and channel.
3. Review frequency, privacy, and permission explanation.
4. Grant permission and confirm.
5. Edit, snooze, or stop the alert later.

Alternate flows

- A1 at step 4: Notification is denied; keep a local saved plan and show manual refresh.
- A2 at step 2: No channel is available; offer in-module status check.

Exception flows

- E1: Service failure preserves unsent preferences for retry.

Field	Record
Business rules	No alert is enabled without affirmative consent; every alert has an unsubscribe control.
Data	Context, selected changes, channel, consent timestamp, retention state.
Usability	Control, prevention of notification fatigue, and clear status.
Accessibility	Permission explanation is plain text; controls are keyboard operable.
Privacy	Minimize contact data and disclose retention.
Traceability	PA1 ticket uncertainty -> F2-E09 -> F-SES-06 -> UR-P-F1 -> UR-F01 -> Freshness and alerts -> 3 votes -> TP-FIFA -> F-A3 -> retained PA3 prototype -> notification-denial test.

2.6 F-UC06 - Continue, stay or return

Field	Record
ID	F-UC06
Goal	Continue, stay or return
Scope	FIFA Ticket Confidence Module
Primary actor	Ticket planner
Supporting actor	Official ticketing partner; identity service
Trigger	User acts from destination preview or reopens a return route.
Preconditions	Comparison context exists; any return token is validated.
Success guarantee	User remains oriented and partner-side completion is never inferred.
Priority	Critical
Frequency	Each handoff and return

Main flow

1. Choose Continue, Stay, or Return.
2. If Continue, transfer only consented context.
3. If Stay, preserve the comparison state.
4. If Return, validate token or identity.
5. Restore context, freshness, and available actions.

Alternate flows

- A1 at step 4: Token expired; offer safe account lookup or plan recreation.
- A2 at step 5: Login mismatch; switch account without exposing stored data.

Exception flows

- E1: Partner or network failure restores the last safe state.

Field	Record
Business rules	A return is not proof of purchase; tokens are time-limited and single-purpose.
Data	Context, return token, account reference, last state and timestamp.
Usability	Recovery, memorability, and persistent orientation.
Accessibility	Restoration is announced without unexpected focus movement.
Privacy	Validate identity before revealing saved plans.
Traceability	PA1 handoff drawback -> F2-E09 -> F-SES-02/03/05/06 -> UR-P-F1/F2 -> UR-F02/F03 -> Recovery and return -> 1 vote -> TP-FIFA -> F-A1 -> PA3 return test.

2.7 C-UC01 - Open game

Field	Record
ID	C-UC01
Goal	Open game
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Game store
Trigger	User chooses a recent game, PGN import, or board setup.
Preconditions	A source route is available; input may be absent or invalid.
Success guarantee	A valid review artifact is opened or input is preserved with a clear correction path.
Priority	High
Frequency	Each review

Main flow

1. Choose Recent game, Import PGN, or Set up position.
2. Select or enter the game artifact.
3. Confirm side to move and board orientation.
4. Validate notation and access rights.
5. Open the review entry.

Alternate flows

- A1 at step 1: No game exists; emphasize import/setup routes.
- A2 at step 4: PGN is invalid; identify the first error and preserve input.

Exception flows

- E1: Mobile interruption preserves the draft locally.

Field	Record
Business rules	Private games follow account permissions; invalid PGN is never silently changed.
Data	Game ID, PGN, ownership, side to move, orientation.
Usability	Clear setup choices and recoverable validation.
Accessibility	Text notation alternative and fully labeled board controls.
Privacy	Respect game visibility and minimize copied metadata.
Traceability	PA1 analysis drawback -> C2-E10 -> C-SES-01/02/05 -> UR-P-C1 -> UR-C01 -> Review setup -> 3 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 setup test.

2.8 C-UC02 - Select beginner review

Field	Record
ID	C-UC02
Goal	Select beginner review
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Analysis service
Trigger	A valid game artifact is opened.
Preconditions	Review entry is visible; premium/access restrictions may apply.
Success guarantee	The beginner preset is selected without removing advanced analysis access.
Priority	Critical
Frequency	Each beginner review

Main flow

1. Open review choices.
2. Read Beginner Review purpose and scope.
3. Select Beginner Review.
4. Confirm the game and perspective.
5. Start the guided review.

Alternate flows

- A1 at step 2: Terminology help is requested; show inline explanation.
- A2 at step 3: Premium restriction applies; explain available scope and alternatives.

Exception flows

- E1: Analysis service unavailable; offer retry and practice from the unreviewed game.

Field	Record
Business rules	Restrictions are stated before commitment; expert controls remain available by choice.
Data	Game artifact, preset, perspective, entitlement.
Usability	Recognition over recall and progressive disclosure.
Accessibility	Keyboard-only selection and focus-visible controls.
Privacy	No extra personal data beyond game access.
Traceability	PA1 analysis drawback -> C2-E10 -> C-SES-01/02/05 -> UR-P-C1 -> UR-C01 -> Entry choice overload -> 4 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 entry-choice test.

2.9 C-UC03 - Identify mistake

Field	Record
ID	C-UC03
Goal	Identify mistake
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Analysis service
Trigger	Guided review completes its first pass.
Preconditions	A review is available; there may be no major mistake or several comparable mistakes.
Success guarantee	One learning priority is shown with honest uncertainty.
Priority	Critical
Frequency	Each review

Main flow

1. Receive the guided review.
2. Show the most important mistake or state that no major mistake exists.
3. Display the position and move context.
4. Explain why this moment matters in plain language.
5. Offer Try a better move.

Alternate flows

- A1 at step 2: No major mistake; show a useful improvement instead.
- A2 at step 2: Multiple mistakes are comparable; explain the selection and allow comparison.

Exception flows

- E1: Engine result is partial; label uncertainty and avoid a definitive claim.

Field	Record
Business rules	One priority does not imply all other moves are irrelevant.
Data	Position, move, evaluation change, confidence, explanation key.
Usability	Effectiveness, clear hierarchy, and calibrated status.
Accessibility	Text description accompanies board highlight; no color-only meaning.
Privacy	Game visibility rules continue to apply.
Traceability	PA1 learning/analysis gap -> C2-E10 -> C-SES-02/04/06 -> UR-P-C1/C2 -> UR-C02 -> Learning priority -> 3 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 mistake-comprehension test.

2.10 C-UC04 - Try better move

Field	Record
ID	C-UC04
Goal	Try better move
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Analysis service
Trigger	User chooses Try a better move.
Preconditions	A review position and legal move set exist.
Success guarantee	The attempted move receives feedback without losing the original position.
Priority	High
Frequency	At each selected mistake

Main flow

1. Open the review position in trial mode.
2. Enter a legal candidate move.
3. Compare it with the learning priority.
4. Receive concise feedback.
5. Retry, reveal the recommended move, or return.

Alternate flows

- A1 at step 2: Move is illegal; explain the rule and keep focus on the board.
- A2 at step 3: Move is also strong; acknowledge it and compare ideas.

Exception flows

- E1: Engine unavailable; record the trial locally and defer comparison.

Field	Record
Business rules	User action in trial mode never changes the stored game.
Data	Position, candidate move, legal moves, comparison result.
Usability	Immediate feedback, user control, and safe experimentation.
Accessibility	Keyboard coordinate entry and announced feedback.
Privacy	Trial data stays with the selected review unless saved.
Traceability	PA1 analysis opportunity -> C2-E10 -> C-SES-02/06 -> UR-P-C1/C2 -> UR-C02 -> Learning priority -> 3 votes -> TP-CHESS -> C-A1 -> PA3 move-trial test.

2.11 C-UC05 - Read explanation

Field	Record
ID	C-UC05
Goal	Read explanation
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Explanation service
Trigger	A mistake or better move is selected.
Preconditions	An explanation exists or a safe fallback can describe the position.
Success guarantee	Plain language leads and terminology help is available on demand.
Priority	Critical
Frequency	Each selected mistake

Main flow

1. Open the explanation card.
2. Read the one-sentence learning point.
3. Review board evidence and move comparison.
4. Open terminology help if needed.
5. Choose Try again, Continue practice, or Advanced detail.

Alternate flows

- A1 at step 2: Explanation is unavailable; show factual board changes and avoid invented rationale.
- A2 at step 4: Terminology help opens inline and returns focus to the term.

Exception flows

- E1: Service failure keeps the move trial and offers retry.

Field	Record
Business rules	Explanations distinguish fact, recommendation, and uncertainty.
Data	Learning point, board evidence, glossary terms, explanation version.
Usability	Plain language, satisfaction, and recognition over recall.
Accessibility	Readable text, glossary semantics, and no forced motion.
Privacy	No unrelated conversation or profile data.
Traceability	PA1 terminology drawback -> C2-E10 -> C-SES-01/02/04/06 -> UR-P-C1/C2 -> UR-C02 -> Vocabulary load -> 5 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 explanation test.

2.12 C-UC06 - Continue to practice or advanced depth

Field	Record
ID	C-UC06
Goal	Continue to practice or advanced depth
Scope	Chess Guided Beginner Review Module
Primary actor	Beginner / returning learner
Supporting actor	Learning catalog; analysis service
Trigger	User completes the explanation.
Preconditions	A learning point exists; practice or depth may be unavailable.
Success guarantee	One relevant practice action is offered before optional advanced depth.
Priority	Critical
Frequency	Each completed explanation

Main flow

1. Review the recommended lesson or puzzle.
2. Read why it matches the mistake.
3. Choose Continue to practice or Reveal advanced detail.
4. Open the chosen destination with review context.
5. Return to the review when desired.

Alternate flows

- A1 at step 1: No exact practice exists; offer a broader topic and explain the gap.
- A2 at step 3: Advanced detail is restricted; explain scope without blocking practice.

Exception flows

- E1: Mobile interruption preserves the review and selected next step.

Field	Record
Business rules	Practice relevance is explainable; advanced depth is opt-in, not removed.
Data	Learning point, catalog item, relevance reason, disclosure state.
Usability	Continuation, memorability, and progressive disclosure.
Accessibility	Destination choice and return route are keyboard/screen-reader operable.
Privacy	Share only the minimum review context with the learning catalog.
Traceability	PA1 learning strength + analysis drawback -> C2-E05/E07/E10 -> C-SES-03/04/06 -> UR-P-C2 -> UR-C03 -> Practice continuation -> 5 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 continuation test.

3. Edge-case coverage

Product	Required edge case	Covered by
FIFA	Stale state; missing state; sold out; resale; waiting room	F-UC02-F-UC03
FIFA	Partner failure; login mismatch; notification denied; network loss	F-UC01, F-UC04-F-UC06
Chess	No game; invalid PGN; engine unavailable	C-UC01-C-UC04
Chess	No major mistake; multiple mistakes; premium restriction	C-UC02-C-UC03
Chess	Mobile interruption; keyboard-only; terminology help	C-UC01-C-UC02, C-UC05-C-UC06

4. Course-source application

Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 - Fundamental Concepts - Usability Dimensions_2.pdf, pp. 62-67, 74-76, 83-84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36; LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

PA3 validation target: Test the six core steps per product and the stated alternate/exception flows without assuming implementation success.